Hide menu

Introduction to KNN Model
Introduction to Decision Tree Model
Decision Tree Metrics
Optimizing Decision Tree Models
Slide Decks
Assessments

Quiz: Week 4 Quiz
Submitted

Programming Assignment: Week 4: Non-parametric Models
3h

Peer-graded Assignment: Week 4: Non-parametric Models
7h

Review Your Peers: Week 4: Non-parametric Models

Peer-graded Assignment: Week 4: Non-parametric Models

You passed!
Congratulations. You earned 30 / 31 points. Review the feedback below and continue the course when you are ready. You can also help more peers by reviewing their submissions.

Review assignments

Instructions | My submission | Discussions

Your fellow learners have submitted their reviews anonymously. All names are still visible to course instructors.

on-parametric Models Project

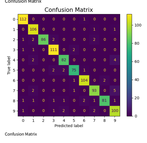
Submitted on November 16, 2023

PROMPT

For Problem 1 of the Module 4 lab, you created a KNN classifier for handwritten digit recognition. Use your KNN class to perform KNN on the validation data with K = 3. Now you will check how well your classifier does. Create a **confusion matrix** in your Jupyter notebook. Feel free to use the Scikit-Learn confusion matrix (http://scikit-learn.org/stable/modules/generated/sklearn.metrics.confusion_matrix.html) function).

Upload a screenshot of the confusion matrix from your notebook here.

Confusion Matrix



Confusion Matrix

RUBRIC

Did the author upload a screenshot of a **confusion matrix** showing the performance of their KNN classifier on the 9 digits in the MNIST dataset?

See an example solution below:

```
confusion matrix:
[[112   0   0   0   0   0   1   0   0   0]
 [  0 106   0   0   0   0   0   1   0   1]
 [  1   2  84   2   0   0   0   2   0   0]
 [  1   1   1 111   0   2   0   0   0   0]
 [  0   2   0   0  82   0   0   0   0   4]
 [  0   0   0   2   2  75   1   0   0   0]
 [  0   0   0   0   0   1 104   0   2   0]
 [  0   2   0   0   0   1   0  93   0   5]
 [  1   1   0   1   1   0   2   1  91   1]
 [  1   1   0   1   2   0   0   2   0 100]]
```

2 points
Learner's response looks reasonable for a confusion matrix showing the performance of their KNN classifier on the 9 digits in the MNIST dataset, e.g. they submitted a **10 x 10 confusion matrix** .

0 points
Learner's response does not look reasonable for a confusion matrix showing the performance of their KNN classifier on the 9 digits in the MNIST dataset, e.g. they **didn't submit a 10 x 10 confusion matrix** .

PROMPT

Based on your confusion matrix, which digits seem to get confused with other digits the most?

Type your answer in the rich text field.

As can be seen from the above confusion matrix plot, the digits 7 and 8 seem to get confused with other digits the most (8 times each).

RUBRIC

Does the author correctly identify which digits are confused with other digits the most?

2 points
Correctly identifies that 9 and 7 are confused with other digits the most.

See how the confusion matrix shows that these two digits were misclassified five times.

```
confusion matrix:
[[112   0   0   0   0   0   1   0   0   0]
 [  0 106   0   0   0   0   0   1   0   1]
 [  1   2  84   2   0   0   0   2   0   0]
 [  1   1   1 111   0   2   0   0   0   0]
 [  0   2   0   0  82   0   0   0   0   4]
 [  0   0   0   2   2  75   1   0   0   0]
 [  0   0   0   0   0   1 104   0   2   0]
 [  0   2   0   0   0   1   0  93   0   5]
 [  1   1   0   1   1   0   2   1  91   1]
 [  1   1   0   1   2   0   0   2   0 100]]
```

1 point
Does not correctly identify that 9 and 7 are confused with other digits the most, but the digits identified are correct based on the confusion matrix that the learner uploaded.

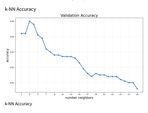
0 points
Learner did not answer the question **or** learner did not identify that 9 and 7 are confused with other digits the most **and** digits the learner identified were not correct based on the confusion matrix that the learner uploaded.

PROMPT

In your Module 4 Jupyter notebook, you created a plot of the accuracy of the KNN on the test set using the same set of axes for K=1, 2, ..., 20. You could go to K=30 if your implementation was efficient enough to allow it.

Please upload a **screenshot of your plot** here.

k-NN Accuracy

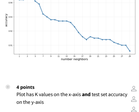


k-NN Accuracy

RUBRIC

Evaluate the learner's KNN accuracy plot.

Here is a sample solution plot:



4 points
Plot has K values on the x-axis **and** test set accuracy on the y-axis

2 points
Plot **missing at least one of the following**: K values on x-axis **or** test set accuracy on the y-axis.

0 points
No plot submitted.

PROMPT

Based on the plot of the KNN accuracy on the test set, which value of K results in highest accuracy?

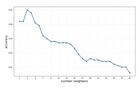
Type your answer in the rich text field.

K = 3 results in the highest accuracy (as can be seen from the above plot and the following list).

```
K  accuracy
1  0.942
2  0.942
3  0.950
4  0.948
5  0.941
6  0.939
7  0.932
8  0.930
9  0.928
10 0.928
11 0.927
12 0.927
13 0.927
14 0.926
15 0.923
16 0.919
17 0.916
18 0.914
19 0.916
20 0.915
21 0.915
22 0.914
23 0.914
24 0.914
25 0.912
26 0.911
27 0.910
28 0.910
29 0.906
```

RUBRIC

From the solution KNN accuracy plot below, we can see that a K value of 3 neighbors results in the highest accuracy.



4 points
Learner correctly identifies that 3 neighbors results in the highest accuracy .

2 points
Learner does not identify a K value of 3 but identifies the highest accuracy K value on the plot that the learner submitted for Prompt 3.

0 points
No answer attempted **or** learner does not identify highest accuracy K value on the plot that the learner submitted.

PROMPT

In Problem 2 about Decision Trees, the directions for Part C1 are to:

1. Modify **max_depth**:
- Create a model with a shallow **max_depth** of 2. Build the model on the training set with the **build_dt** function.
- Report **precision/recall** on the test set.
- Report **depth of the tree.**

Upload a screenshot of your code for this section.

DT Precision Recall

```
# In Problem 2 about Decision Trees, the directions for Part C1 are to:
# 1. Modify max_depth:
#    - Create a model with a shallow max_depth of 2. Build the model on the training set with the build_dt function.
#    - Report precision/recall on the test set.
#    - Report depth of the tree.
# dt = build_dt(X_train, y_train, max_depth = 2)
# y_test_hat = dt.predict(X_test)
# precision = calculate_precision(y_test, y_test_hat)
# recall = calculate_recall(y_test, y_test_hat)
# print("Precision: {:0.2f} Recall: {:0.2f}".format(precision, recall))
# # Report depth of the tree.
# print(dt.get_depth())
```

DT Precision Recall

RUBRIC

Here is a **solution**:

```
dt = build_dt(X_train, y_train, max_depth= 2)

y_test_hat = dt.predict(X_test)


prediction = calculate_precision(y_test, y_test_hat)

recall = calculate_recall(y_test, y_test_hat)
print("Precision: {:0.2f} Recall: {:0.2f} Tree Depth: {}".format(prediction, recall, dt.get_depth()))
```

Evaluate the learner's submission.

5 points
Includes all of the following: Learner calls build_dt function with max_depth=2 **and** model built on the training set **and** calculates precision on the test set **and** calculates recall on the test set **and** reports depth of the tree.

4 points
Missing one of the following: Learner calls build_dt function with max_depth=2 **or** model built on the training set or calculates precision on the test set **or** calculates recall on the test set **or** reports depth of the tree.

3 points
Missing two of the following: Learner calls build_dt function with max_depth=2 **or** model built on the training set or calculates precision on the test set **or** calculates recall on the test set **or** reports depth of the tree.

2 points
Missing three of the following: Learner calls build_dt function with max_depth=2 **or** model built on the training set or calculates precision on the test set **or** calculates recall on the test set **or** reports depth of the tree.

1 point
Missing four of the following: Learner calls build_dt function with max_depth=2 **or** model built on the training set or calculates precision on the test set **or** calculates recall on the test set **or** reports depth of the tree.

0 points
Missing all of the following: Learner calls build_dt function with max_depth=2 **and** model built on the training set **and** calculates precision on the test set **and** calculates recall on the test set **and** reports depth of the tree.

PROMPT

In Problem 2 about Decision Trees, the directions for Part C1 are to:

2. Modifying **max_leaf_nodes**:
Create a model with a shallow **max_leaf_nodes of 4**. Build the model on the training set.

Upload a screenshot of your code for this section.

DT Precision Recall

```
# 2. Modifying max_leaf_nodes:
#    Create a model with a shallow max_leaf_nodes of 4. Build the model on the training set.
# dt = build_dt(X_train, y_train, max_leaf_nodes = 4)
# y_test_hat = dt.predict(X_test)
# # Report precision/recall on the test set
# precision = calculate_precision(y_test, y_test_hat)
# recall = calculate_recall(y_test, y_test_hat)
# # Report depth of the tree.
# print("Precision: {:0.2f} Recall: {:0.2f}".format(precision, recall))
# print(dt.get_depth())
```

DT Precision Recall

RUBRIC

Here is a solution:

```
dt = build_dt(X_train, y_train, max_depth =None, max_leaf_nodes = 4)

y_test_hat = dt.predict(X_test)


precision = calculate_precision(y_test, y_test_hat)

recall = calculate_recall(y_test, y_test_hat)

print("Precision: {:0.2f} Recall: {:0.2f} No. of leaves: {}".format(precision, recall, dt.get_n_leaves()))
```

Evaluate the learner's solution.

5 points
Includes all of the following: Learner calls build_dt function with max_leaf_nodes = 4 **and** model built on the training set **and** calculates precision on the test set **and** calculates recall on the test set **and** reports depth of the tree.

4 points
Missing one of the following: Learner calls build_dt function with max_leaf_nodes = 4 **or** model built on the training set or calculates precision on the test set **or** calculates recall on the test set **or** reports depth of the tree.

3 points
Missing two of the following: Learner calls build_dt function with max_leaf_nodes = 4 **or** model built on the training set or calculates precision on the test set **or** calculates recall on the test set **or** reports depth of the tree.

2 points
Missing three of the following: Learner calls build_dt function with max_leaf_nodes = 4 **or** model built on the training set or calculates precision on the test set **or** calculates recall on the test set **or** reports depth of the tree.

1 point
Missing two of the following: Learner calls build_dt function with max_leaf_nodes = 4 **or** model built on the training set or calculates precision on the test set **or** calculates recall on the test set **or** reports depth of the tree.

0 points
Missing all of the following: Learner calls build_dt function with max_leaf_nodes = 4 **and** model built on the training set **and** calculates precision on the test set **and** calculates recall on the test set **and** reports depth of the tree.

PROMPT

The final part of question C1 refers to your code in the previous two prompts:

How do precision and recall compare when you modify the max depth compared to the max number of leaf nodes?
(Make sure to run your models a few times to get an idea).

Type your answer in the rich text field.

1. When the max depth is modified, the **precision** of the model is **high** and the **recall** is **low** (on the held-out validation dataset).

2. When the max number of leaf nodes is modified, the **precision** of the model is **low** and the **recall** is **high** (on the held-out validation dataset).

RUBRIC

Evaluate whether the learner answers the question in a reasonable way.

2 points
Reasonable answer comparing precision and recall when modifying the max depth compared to modifying the max number of leaf nodes. Answer seems to be supported by running models with the modifications.

1 point
Answer is reasonable and seems to be supported by running models with the modifications but is partially missing - e.g., only refers to precision but not recall.

0 points
Answer is not a comparison of precision and recall when modifying the max depth compared to modifying the max number of leaf nodes.

PROMPT

In Problem 2 about Decision Trees, the directions for Part D are as follows:

In class, we used gridsearchCV to do hyperparameter tuning to select the different parameters like **max_depth** to see how our tree grows and avoids overfitting. Here, we will use cost complexity pruning parameter **alpha** (sklearn 0.22.1(https://scikit-learn.org/stable/user_guide.html) to prune our tree after training so as to improve accuracy on unseen data. In this exercise you will iterate over different **ccp_alpha** values and identify how performance is modulated by this parameter.

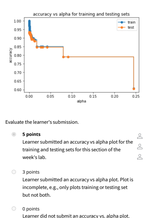
Please **upload a screenshot or copy of your accuracy vs. alpha plot for the training and testing sets.**

Accuracy vs. Alpha Plot

Accuracy vs. Alpha Plot

RUBRIC

Here is a solution plot:



Evaluate the learner's submission.

5 points
Learner submitted an accuracy vs alpha plot for the training and testing sets for this section of the week's lab.

3 points
Learner submitted an accuracy vs alpha plot. Plot is incomplete, e.g., only plots training or testing set but not both.

0 points
Learner did not submit an accuracy vs. alpha plot.

PROMPT

Based on your plot for the prompt above, Problem 2D:

How is performance modulated by this parameter?

Type your answer in the rich text field.

1. With **low** values of the pruning parameter **alpha**, the decision tree model clearly **overfits** (accuracy is high on training but low on unseen test data).

2. As **alpha** value is gradually **increased** from the low value, the **overfitting decreases** and the decision tree model gets more **generalized** (accuracies on training and test dataset become similar), but both the accuracy values **decrease** with increase in **alpha**.

3. When the **alpha** value becomes **high**, the model **underfits** (both the training and test accuracy become poor).

RUBRIC

Evaluate whether the learner's response is reasonable based on their plot.

2 points
Learner's answer is knowledgeable about the ccp_alpha parameter and makes sense based on the plot that they submitted.

1 point
Learner's answer **missing one of the following**: learner's answer is knowledgeable about the ccp_alpha parameter **or** makes sense based on the plot that they submitted.

0 points
Learner's answer **missing both of the following**: learner's answer is knowledgeable about the ccp_alpha parameter **and** makes sense based on the plot that they submitted.

Start new attempt

Comments
Comments left for the learner are visible only to that learner and the person who left the comment.

SD Share your thoughts...

Like Dislike Report an issue